Machine Learning vs. Deep Learning

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I am still learning how to tell machine learning and deep learning apart because deep learning is still a type of machine learning, so I am taking this as an opportunity to explore both. I will compare actual examples to help me understand what makes them different and why it's better to choose one over the other depending on the use case. I chose predicting house prices with linear regression and identifying plant diseases with a convolutional neural network (CNN). I understand that both learn from examples, but the information they work with is different.

Predicting House Prices

Pardoe (2008) used realtor data from Oregon to model home prices through multiple linear regression. The data analysis took in characteristics like floor size, age, bedrooms, and bathrooms. The inputs describe things we would consider when comparing houses, so the model learns how those characteristics relate to housing prices.

I would choose traditional machine learning here because the information is organized into specific variables and can be used to explain a certain decision or results. A buyer or broker might want to understand which variables are associated with a higher price. However, Pardoe (2008) also showed that there is still a significant amount of uncertainty in this situation.

I learned that using deep learning on a small dataset is not useful because the model could learn the training examples too thoroughly but still make inaccurate predictions for other houses. A more complicated model could fit the available examples without predicting new ones well. To use deep learning for this situation, I would need to find evidence that it predicts better and measure the accuracy of predictions to justify adding complexity.

Recognizing Plant Diseases

Ramcharan et al. (2017) used photographs taken in Tanzania to identify diseases and pest damage in plants. Their system used a trained CNN to learn useful patterns for the task. This example is an application of image classification and here the categories describe conditions of a plant.

The model needs to identify patterns that can identify what is happening with a plant, for example is a disease progresses. The identifiers can be visible damage, discoloration on leaves, which all needs to be identified by the model and categorized correctly. Deep learning would be good to use here because a CNN can learn to identify visual features against various backgrounds without being given direct guidelines on what to look for.

A traditional model could still classify these images if someone first identified and extracted useful features, but then the issue would be to decide which features capture the important differences. I learned that the study actually combined CNN features with different classifiers, including an SVM (Ramcharan et al., 2017), so turns out that traditional ML and deep learning can also work together.

At the end, I would not assume that either deep learning or traditional ML is better and the choice depends on what the problem requirements are.

References

Pardoe, I. (2008). Modeling home prices using realtor data. *Journal of Statistics Education, 16*(2). https://doi.org/10.1080/10691898.2008.11889569

Ramcharan, A., Baranowski, K., McCloskey, P., Ahmed, B., Legg, J., & Hughes, D. P. (2017). Deep learning for image-based cassava disease detection. *Frontiers in Plant Science, 8*, Article 1852. https://doi.org/10.3389/fpls.2017.01852

AI Use Disclosure

I used ChatGPT (OpenAI) to help locate research articles.